

Quantitative Methods for Business and Economics

Lecture 2 - Sigma Notation and Further Topics on Algebra

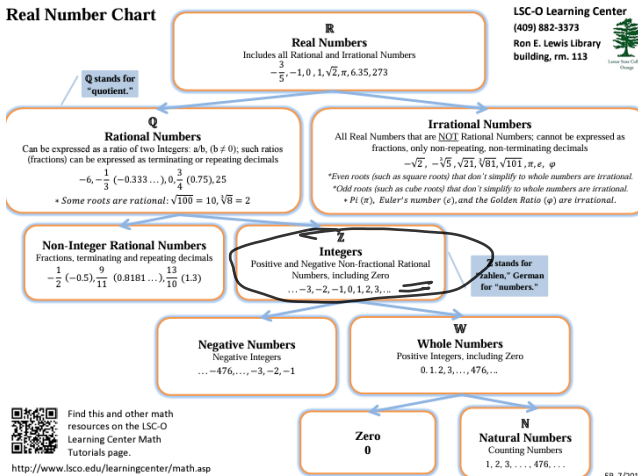
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2026

A quick recap on real numbers

- The set of real numbers is denoted by $\mathbb{R}$.

Real Number Chart



EP, 7/2013

Summation

- In some cases we will be interested in the sum of a sequence of numbers.
- For example, the sum of the first 10 natural numbers is given by:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

- What if we want to sum the first 100 natural numbers?
 - It is a lot to write down! Let's use an *ellipsis* to denote the sequence.

$$1 + 2 + 3 + \dots + 98 + 99 + 100 = 5,050$$

- However, this is still a lot to write down!

Sigma or summation notation

Σ

- We can use the Greek letter Σ (capital sigma) to denote the sum of a sequence of numbers.
- The sum of the first 10 natural numbers can be written as:

$$\sum_{i=1}^{10} i = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$$

Handwritten notes: An arrow points from the index i to $i \in \mathbb{Z}$. A bracket underneath the sum points to the right.

- Notice that the index i represents the starting point of the sum.
 - The upper number denotes the last number to be added.

$$\sum_{i=2}^{24} i = 2 + 3 + \dots + 24 = m$$

Sigma or summation notation

- More generally, the sum of the first n natural numbers can be written as:

$$\sum_{i=1}^n i = 1 + 2 + 3 + \dots + \underline{\underline{n}}$$

endpoint (pointing to n)

- In other situations, i simply represents the index of the summands. For example:

$$\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n \quad \underline{\underline{n^{th}}}$$

This would be the sum of any sequence of numbers $a_1, a_2, a_3, \dots, a_n$.

$$\underline{\underline{1 + 2 + 3 \quad 100 + 101 + \dots}}$$

Averages and sums

- The arithmetic mean (or average) is the sum of a group of numbers, divided by the count of numbers. For example:

$$\text{Average} = \frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10}{10} = 5.5$$

- More generally, any average can be written as:

$\bar{x}$ → average
(arithmetic mean)

$$\bar{x} = \frac{1}{n} \left(\sum_{i=1}^n x_i \right)$$

↳ the count of numbers...

$$x_1 = 1.0 \text{ m}$$

$$x_2 = 1.7 \text{ m} \dots$$

where x_i are the numbers in the sequence and n is the count of numbers.

$$= x_1 = 1 ; x_2 = 2 ; \dots ; x_{10} = 10$$

Properties of sums

- The sum of a constant times a sequence of numbers is equal to the constant times the sum of the sequence.

$$\sum_{i=1}^n c x_i = c \sum_{i=1}^n x_i \rightarrow \sum_{i=1}^{10} 3 x_i = 3(1) + 3(2) + \dots + 3(10)$$

This is a re-statement of the distributive property of addition. For example, if $c = 2$:

$$= 3(1+2+\dots)$$


$$2(1 + 2 + 3 + 4 + 5) = 2 + 4 + 6 + 8 + 10 = 30$$

Properties of sums

- The sum of two sequences of numbers is equal to the sum of the individual sequences.

$$\sum_{i=1}^n x_i + \sum_{i=1}^n y_i = \sum_{i=1}^n (x_i + y_i)$$

Handwritten notes and diagrams illustrating the property:

- A bracket under the left side of the equation groups the two sums.
- A bracket under the right side of the equation groups the terms $x_i + y_i$.
- Handwritten arrows point from the underlined text "individual sequences" to the two separate sums on the left.
- Handwritten notes to the right show the summation indices: $\sum_{i=1}^n x_i + \sum_{i=1}^n y_i$ and $\sum_{i=1}^n (x_i + y_i)$.

- This is a re-statement of the associative property of addition. See below:

$$(1 + 2) + (3 + 4) = 1 + 2 + 3 + 4 = 10$$

Handwritten bracket under the right side of the equation, indicating the sum of all terms.

Exponents

- Exponents are a shorthand notation for repeated multiplication.

$2^3 = 2 \cdot 2 \cdot 2 = 8$
 exponent $\swarrow$
 $x^n = x \cdot x \cdot x \cdot \dots \cdot x$ (n times)
 $\swarrow$
 base

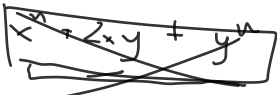
- Properties of exponents:

$\left[\begin{array}{l} \blacksquare x^0 = 1 \text{ for any } x \neq 0. \\ \blacksquare x^1 = x \text{ for any } x. \\ \blacksquare x^m \cdot x^n = x^{m+n}. \\ \blacksquare (x^m)^n = x^{m \cdot n}. \\ \blacksquare (x \cdot y)^n = x^n \cdot y^n. \\ \blacksquare \frac{x^m}{x^n} = x^{m-n}. \\ \blacksquare x^{-n} = \frac{1}{x^n}. \end{array} \right] \rightarrow \text{base is the same}$

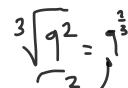
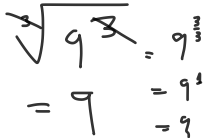
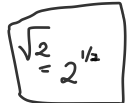
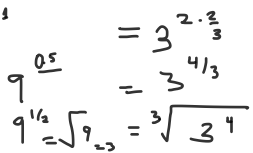
$2^{-1} = \frac{1}{2}$

$2^3 \cdot 2^2 = 2^{3+2} = 2^5$

Common mistakes with exponents

- $x^m \cdot y^n = x^{m+n}$ $\rightarrow$ same base to do this!
- $(x + y)^n = x^n + y^n$ $\rightarrow$ 
- $(x^m)^n = x^{m+n}$ $\rightarrow$ not sum!

Radicals

- The n -th root of a number x is denoted by $\sqrt[n]{x}$. 
- Most common roots are the square root ($n = 2$) and the cube root ($n = 3$). 
- In English, you may hear the square root of x as “the sqrt” of x . 
 - ~~This might be confusing in Spanish, where “la raíz” is used.~~
 - Emerges from a coding context in many mathematical/statistical software: sqrt(x).
 - A calculator will typically understand sqrt(4) as 2, and so on. 
- Properties of radicals:
 - $\sqrt[n]{x} \cdot \sqrt[n]{y} = \sqrt[n]{x \cdot y}$. 
 - $\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$
 - $\sqrt[n]{x^m} = x^{\frac{m}{n}}$. 
 - $\sqrt[n]{x^n} = x$. 

Common mistakes with radicals

- eqn's
- When faced with squared variables, it is common to think that the square root of x^2 is x . For example:

✗

$$\begin{aligned}x^2 &= 4 \\ \sqrt{x^2} &= \sqrt{4} \\ x &= 2\end{aligned}$$

$$(2)(2) = 4$$

$$(-2)(-2) = 4$$

- Stating that only $x = 2$ is a mistake. The correct answer is $x = \pm 2$.
This is since both 2 and -2 squared are equal to 4. ± 2
- To avoid this mistake, always attach the $\pm$ sign when taking the square root of a squared variable in an equation.

Quadratic equations

- A quadratic equation is a polynomial equation of the form:

$$ax^2 + bx + c = 0$$

a, b, c → numbers
constant

- For example, the equation $3x^2 - 2x - 1 = 0$ is a quadratic equation.
- If you get $x^2 = 4$, you have that $a = 1$, $b = 0$, and $c = -4$.
- We are often interested in finding the solutions to a quadratic equation. These are the values of x that make the equation true.
- Also called the roots of the equation, or the zeros of the equation.

$$\begin{aligned} x &= 2 \\ x &= -2 \end{aligned}$$

$$x^2 = 4$$

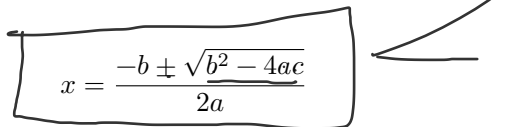
$$x^2 - 4 = 0$$



$$\begin{aligned} 1x^2 + 0x - 4 &= 0 \\ x^2 - 4 &= 0 \end{aligned}$$

Zeros of a quadratic equation

- The solutions to a quadratic equation can be found using the quadratic formula:


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Alternatively, you can factor the quadratic equation and solve for x .
- For example, the equation $x^2 - 5x + 6 = 0$ can be factored as:


$$(x - 2)(x - 3) = 0$$

- Take each factor $(x - 2)$ and $(x - 3)$ and set them equal to zero
 - This is often necessary when the quadratic formula is not practical.
 - You will need to do this often in optimization problems.
- Taking square roots of both sides might work, but only if the equation allows you to do so, and you're careful with the $\pm$ sign.

When are solutions not real?

$$x^2 + 5x + 0 = 0$$
$$ax^2 + bx + c$$

$$a = 1$$
$$b = 5$$
$$c = 0$$

$$b^2 - 4ac$$

$$5^2 - 4(1)(0)$$

$$5^2 - 0 = \underline{\underline{25}}$$

- Sometimes the solutions to a quadratic equation are not real numbers.

- This happens when the discriminant $b^2 - 4ac$ is negative.

- For example, the equation $x^2 + 1 = 0$ has no real solutions.

- For our purposes, we will not be interested in complex numbers.

$$x^2 + 1 = 0$$
$$ax^2 + bx + c$$

$$b^2 - 4ac$$

$$0^2 - 4(1)(1) = -4$$

$$a = 1$$

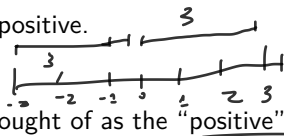
$$c = 1$$

$$b = 0$$

no
soln

Absolute value

- The absolute value of a number x is denoted by $|x|$.
- The absolute value of a number is the distance of the number from zero on the number line.
- The absolute value of a number is always positive.
- For example, $|3| = 3$ and $|-3| = 3$.
- The absolute value of a number can be thought of as the "positive" version of the number.



official



Properties of absolute value

- 
- The absolute value of a number is zero if and only if the number is zero.
 - The absolute value of a number is positive if the number is not zero.
 - The absolute value of a number is the same as the number if the number is positive.
 - The absolute value of a number is the negative of the number if the number is negative.
 - The absolute value of a sum is less than or equal to the sum of the absolute values.

$$|x + y| \leq |x| + |y|$$

Application: break-even analysis



- A firm sells a product at price P per unit
- **Fixed costs** FC : rent, salaries, equipment — paid regardless of output
- **Variable cost** VC : materials, labour per unit — scales with quantity Q
- Total revenue: $TR = P \cdot Q$; Total cost: $TC = FC + (VC \cdot Q)$ ^{per unit}
- The firm **breaks even** when it neither profits nor loses money:
 $TR = TC$

$$\begin{aligned} TR &= P \cdot Q \\ &= 10 \cdot 5 = 50 \end{aligned}$$

Application: break-even analysis

- Set $TR = TC$ and solve for Q :

$TR = TC$

$P \cdot Q = FC + VC \cdot Q$

isolate Q

$P \cdot Q = FC + VC \cdot Q$ $\rightarrow$ expression

$(P \cdot Q) - (VC \cdot Q) = FC$

$(P - VC) \cdot Q = FC$

$Q^* = \frac{FC}{P - VC}$ $\rightarrow$ break even

$Q \rightarrow$ is the break even.

- $(P - VC)$ is the **contribution margin**: how much each unit sold contributes toward covering fixed costs (profit)
- Once Q^* units are sold, every additional unit generates profit

Example: break-even analysis

- A firm has monthly fixed costs of \$10,000, a variable cost of \$6 per unit, and sells at \$16 per unit

Handwritten notes: "rent" (above FC), "materials" (above VC), "price" (below price), "I make profit" (next to the final result).

$$Q^* = \frac{10,000}{16 - 6} = \frac{10,000}{10} = 1,000 \text{ units}$$

Handwritten calculations: $FC = 10,000$, $VC = 6$, $P = 16$, and $\rightarrow 1,000$ (with a plus sign).

- The firm must sell 1,000 units per month to break even
- For you: a bakery has monthly fixed costs of \$3,000, makes each cake for \$8, and sells it for \$20. How many cakes must they sell to break even?

$$Q = \frac{3,000}{20 - 8} = \frac{3,000}{12} = 250$$